

Topics:

- Subspaces: Axler 1.C, Strang 3.1
- Span, bases, and dimension: Axler §2, Strang 3.4

Column and Row vectors

For $\vec{x} \in \mathbb{R}^n$ we express $\vec{x}$ as a column vector

$$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

and the associated row vector as

$$\vec{x}^T = [x_1, \dots, x_n]$$

Now the inner product on $\mathbb{R}^n$ can be written as

$$\langle \vec{x}, \vec{y} \rangle = \vec{x}^T \vec{y} = [x_1, \dots, x_n] \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \sum_{i=1}^n x_i y_i$$

Exercise:

1) let $\vec{c} \in \mathbb{R}^3$ and $b \in \mathbb{R}$ be fixed and describe the following regions of $\mathbb{R}^3$

a. $\vec{c}^T \vec{x} = b \rightarrow c_1 x_1 + c_2 x_2 + c_3 x_3 = b$
plane orthogonal to $\vec{c}$

b. $\vec{x}^T \vec{x} = b^2$ sphere of radius $|b|$
centered at the origin

Subspaces

subset

Def 2.1 let V be a vector space. Then $W \subseteq V$ is a subspace (of V) iff W itself is a vector space (over the same field as V)

Theorem 2.2 $W \subseteq V$ is a subspace iff

i. $\vec{0} \in W$

ii. $\lambda_1 \vec{x}_1 + \lambda_2 \vec{x}_2 \in W$ whenever $\vec{x}_1, \vec{x}_2 \in W$ and $\lambda_1, \lambda_2 \in F$. (closed under linear comb.)

Def 2.3 If $W = \{\vec{0}\}$ then we say that W is trivial and denote $W = \phi$. We say that W is a proper subspace of V if $W \neq \phi$ and $W \neq V$.

The proper subspaces of $\mathbb{R}^3$ are lines and planes through the origin.

Exercise

2) Prove that the following are subspaces of $\mathbb{R}^n$

a. $W_1 = \{ \vec{x} \in \mathbb{R}^n : \langle \vec{x}, \vec{c} \rangle = 0 \}$ for fixed $\vec{c} \in \mathbb{R}^n$

b. $W_2 = \{ \vec{x} \in \mathbb{R}^n : \langle \vec{x}, \vec{y} \rangle = 0 \text{ for all } \vec{y} \in W_1 \}$

Solutions:

a. Since $\langle \vec{0}, \vec{c} \rangle = 0$ we have that $\vec{0} \in W_1$.

Now let $\vec{x}_1, \vec{x}_2 \in W_1$ so that

$$\langle \vec{x}_1, \vec{c} \rangle = \langle \vec{x}_2, \vec{c} \rangle = 0$$

Now as

$$\begin{aligned} \langle \lambda_1 \vec{x}_1 + \lambda_2 \vec{x}_2, \vec{c} \rangle &= \lambda_1 \langle \vec{x}_1, \vec{c} \rangle + \lambda_2 \langle \vec{x}_2, \vec{c} \rangle \\ &= \lambda_1 \cdot 0 + \lambda_2 \cdot 0 \\ &= 0 \end{aligned}$$

we have $\lambda_1 \vec{x}_1 + \lambda_2 \vec{x}_2 \in W_1$ as desired.

b. similar

Algebra of Subspace

Def 2.4 Let U_1 and U_2 be subspaces of V

- i. $U_1 \cup U_2 = \{ \vec{x} \in V : \vec{x} \in U_1 \text{ or } \vec{x} \in U_2 \}$
- ii. $U_1 \cap U_2 = \{ \vec{x} \in V : \vec{x} \in U_1 \text{ and } \vec{x} \in U_2 \}$
- iii. $U_1 + U_2 = \{ \vec{x} \in V : \vec{x} = \vec{u}_1 + \vec{u}_2 \text{ for some } \vec{u}_1 \in U_1 \text{ and } \vec{u}_2 \in U_2 \}$

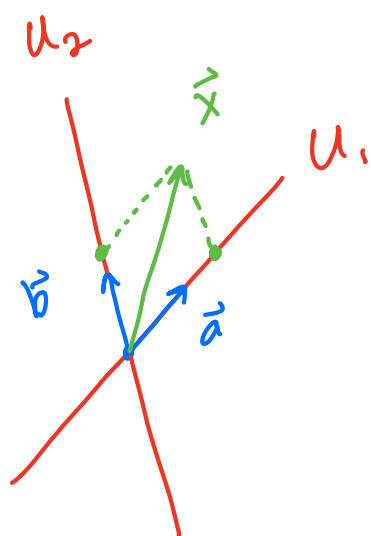
If for every $\vec{x} \in U_1 + U_2$, there exists unique $\vec{u}_1 \in U_1$ and $\vec{u}_2 \in U_2$ s.t. $\vec{x} = \vec{u}_1 + \vec{u}_2$ then we say that $U_1 + U_2$ is the direct sum $U_1 \oplus U_2$.

Theorem 2.5 If $U_1 \cap U_2 = \{ \vec{0} \}$ (only contains the zero vector) then $U_1 + U_2 = U_1 \oplus U_2$.

Proof: Exercise

Exercise:

- 3) Let $\vec{a}$ and $\vec{b}$ be non zero vectors in $\mathbb{R}^3$ s.t. $\vec{a} \neq \vec{b}$. Setting $U_1 = \{ s\vec{a} : s \in \mathbb{R} \}$ and $U_2 = \{ t\vec{b} : t \in \mathbb{R} \}$, describe $U_1 + U_2$.



$U_1 + U_2 = \begin{cases} \text{plane spanned by } \vec{a} \text{ and } \vec{b} \\ \text{if } \vec{a} + \lambda \vec{b} \neq \vec{0} \text{ for all } \lambda \in \mathbb{R} \\ U_1 \oplus U_2 \\ \text{line spanned by } \vec{a} \text{ (or } \vec{b}) \\ \text{if } \vec{a} + \lambda \vec{b} = \vec{0} \text{ for some } \lambda \in \mathbb{R} \\ U_1 \text{ or } U_2 \end{cases}$

Linear Span, Bases, Dimension

Def 2.6 Let V be a vector space over F and $\vec{v}_i \in V$ for $i=1, \dots, n$. The span of the set of vectors $\{\vec{v}_1, \dots, \vec{v}_n\}$ is the set of all linear combinations denoted by

$$\text{Span}(\vec{v}_1, \dots, \vec{v}_n) = \left\{ \sum_{i=1}^n c_i \vec{v}_i : c_i \in F \right\}$$

Exercise:

4) Describe the linear spans over $\mathbb{R}$

a. $\text{Span} \left(\begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \right)$

The plane through the origin containing

these vectors are normal to
 $(3, 1, 0) \times (2, 0, -1)$

$$b. \text{Span}(x, x^2, x^3) = \{ax + bx^2 + cx^3 : a, b, c \in \mathbb{R}\}$$

set of cubics s.t. $f(0) = 0$

Theorem 2.7 $\text{Span}(\vec{v}_1, \dots, \vec{v}_n)$ is the smallest subspace containing the above vectors

Let W be a subspace s.t. $\vec{v}_i \in W$ for $1 \leq i \leq n$. Then $\text{Span}(\vec{v}_1, \dots, \vec{v}_n)$ is a subspace of W .

Linear Independence

Def 2.8 A set of non zero vectors $\{\vec{v}_1, \dots, \vec{v}_n\}$ is linearly independent iff $\sum_{i=1}^n c_i \vec{v}_i = \vec{0}$ implies that $c_i = 0$ for each i . A set of vectors is linearly dependent if they are not linearly independent.

Theorem 2.9 A set of vectors $\{\vec{v}_1, \dots, \vec{v}_n\}$ is linearly dependent iff there exists $1 \leq j \leq n$ s.t. $\vec{v}_j = \sum_{i \neq j} c_i \vec{v}_i$. I.e. $\vec{v}_j \in \text{Span}(\vec{v}_i, i \neq j)$

Exercise :

5) Show that the following vectors are linearly dependent and describe their span

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} -2 \\ 0 \\ -2 \end{bmatrix}, \quad \vec{v}_3 = \begin{bmatrix} 5 \\ -3 \\ 2 \end{bmatrix}$$

Solution 1: Set $\vec{v}_3 = c_1 \vec{v}_1 + c_2 \vec{v}_2$ to get

$$\begin{bmatrix} 1 & -2 \\ 3 & 0 \\ 4 & -2 \end{bmatrix} \vec{c} = \begin{bmatrix} 5 \\ -3 \\ 2 \end{bmatrix}$$

and eliminating

$$\left[\begin{array}{cc|c} 1 & -2 & 5 \\ 3 & 0 & -3 \\ 4 & -2 & 2 \end{array} \right] \xrightarrow{-r_1} \left[\begin{array}{cc|c} 1 & -2 & 5 \\ 3 & 0 & -3 \\ 3 & 0 & -3 \end{array} \right]$$

and thus $c_1 = -1$ and $c_2 = -3$. Now as $\vec{v}_3 = -\vec{v}_1 - 3\vec{v}_2$ the set of vectors is lin. dep.

Solution 2: Since 3rd component is the sum of the first two we have that

$$\vec{v}_i \in \left\{ (x_1, x_2, x_3) \in \mathbb{R}^3 : \overset{x_1 + x_2 - x_3 = 0}{x_3 = x_1 + x_2} \right\}$$

which is a plane through the origin and

normal to $(1,1,-1)$ and is thus has dimension 2. Thus the set $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is lin. dep. Here

$$\text{Span}(\vec{v}_1, \vec{v}_2, \vec{v}_3) = \text{Span}(\vec{v}_1, \vec{v}_2)$$

Bases

Def 2.10 $\mathcal{B} \subseteq V$ is a basis of V iff it is linearly independent and $V = \text{Span } \mathcal{B}$

(technically $V = \overline{\text{Span } \mathcal{B}}$ if $\mathcal{B}$ is infinite)
(closure)

* Bases are minimal spanning sets or maximal linearly independent set.

Theorem 2.11 Let $\mathcal{B}$ be a basis for V s.t. $|\mathcal{B}| = n$. Then for any other basis $\mathcal{B}'$ of V , $|\mathcal{B}'| = |\mathcal{B}| = n$.

Main idea: Pick two bases $\mathcal{B} = \{\vec{v}_1, \dots, \vec{v}_n\}$ and $\mathcal{B}' = \{\vec{w}_1, \dots, \vec{w}_m\}$ and show that $n = m$ by showing that $n \leq m$ and $n \geq m$.

Def 2.12 If $\mathcal{B}$ is a basis for V with $|\mathcal{B}| = n$

then the dimension of V denoted $\dim V$ is n
($\dim V = n$)

Examples: $\dim \mathbb{R}^n = n$, $\dim \mathbb{P}_n(F) = n+1$

Recall that the standard basis for $\mathbb{R}^n$ is

$$\{\vec{e}_1, \dots, \vec{e}_n\} \text{ where } \vec{e}_i = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}_i$$

$$\text{in } \mathbb{R}^3: \vec{e}_1 = \hat{i}, \vec{e}_2 = \hat{j}, \vec{e}_3 = \hat{k}$$

Theorem 2.13 let V be finite dimensional
and let U_1 and U_2 be subspaces

- i. $\dim U_i \leq \dim V$
- ii. $\dim (U_1 + U_2) = \dim U_1 + \dim U_2 - \dim (U_1 \cap U_2)$
- iii. $\dim (U_1 \oplus U_2) = \dim U_1 + \dim U_2$

Main idea: For basis $\mathcal{B}_i$ of subspace U_i
we can construct basis $\mathcal{B}$ of V s.t. $\mathcal{B}_i \subseteq \mathcal{B}$

Polynomial space $\mathbb{P}_n$

Def 2.14 $\mathbb{P}_n(\mathbb{R}) = \left\{ \sum_{i=0}^n a_i x^i : a_i \in \mathbb{R} \right\}$

$P_n(\mathbb{R})$ is a vector space of dim $n+1$ with standard basis $\{1, x, x^2, \dots, x^n\}$ $e_j = x^{j-1}$

We can turn $P_n(\mathbb{R})$ into an inner product space by $\langle \cdot, \cdot \rangle : P_n \times P_n \rightarrow \mathbb{R}$ defined as

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx$$

This admits the L^2 norm (over $\mathbb{R}$)

$$\|f\|_2 = \sqrt{\langle f, f \rangle} = \left(\int_0^1 f(x)^2 dx \right)^{1/2}$$